

Week 5 Project Summary — Data Analytics Track

AnalystLab Africa — HealthConnect Experience Lab

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1. What I Planned to Accomplish

The main goal for Week 5 was to move from the Week 4 planning and initial assessment stage into practical analysis. I planned to validate the HealthConnect appointment data, perform exploratory analysis, calculate and interpret the proposed KPIs, identify meaningful business insights, and develop an initial Power BI analytical dashboard.

This followed the Week 4 approach of using descriptive and correlational analysis to understand appointment No-Show patterns.

2. What I Actually Completed

During Week 5, I:

- Loaded and prepared the HealthConnect appointment dataset using Python and Pandas.
- Reviewed data types, missing values, duplicates, and categorical consistency.
- Validated important data relationships and constraints.
- Performed exploratory analysis across reminders, booking lead time, previous No-Show history, distance, appointment type, and gender.
- Calculated and interpreted the five selected KPIs.
- Performed a combined analysis of booking lead time and previous No-Show history.
- Developed business insights and practical recommendations.
- Documented the limitations and considerations of the analysis.
- Built an initial Power BI dashboard containing KPI cards, analytical visuals, and interactive slicers.

3. Key Findings and Development Outcomes

The analysis identified several clear patterns in the dataset:

- Booking lead time showed the largest observed difference in No-Show rates, increasing from 27.8% for appointments booked within the same week to 56.9% for appointments booked 3+ weeks in advance.
- Patients with previous No-Show history had a higher observed No-Show rate (55.4%) than those without previous No-Shows (43.5%).
- The combination of 3+ week booking lead time and previous No-Show history had the highest observed No-Show rate at 64.1%.
- Appointments 15km+ from the clinic had a higher observed No-Show rate (54.1%) than appointments less than 5km away (46.5%).
- Reminder status showed a smaller difference, with 47.4% No-Show among appointments with reminders compared with 51.4% without reminders.
- Appointment type and gender showed relatively limited differences in No-Show rates.

The Power BI dashboard was developed to communicate these patterns clearly and allow further exploration through Appointment Type and Gender slicers.

4. Major Challenges Encountered

The main challenges were:

- Handling structural missing values in `reminder\_channel`, where missing values corresponded to appointments for which no reminder was sent.
- Deciding how to communicate findings without presenting observed relationships as causal effects.
- Translating the Python analysis into appropriate Power BI measures and visuals.
- Ensuring the Power BI dashboard results matched the calculations produced in Python.
- Choosing appropriate groupings for variables such as booking lead time and distance so that the results remained meaningful and easy to interpret.
- Encountering a persistent circular dependency error in Power BI when attempting to apply a custom chronological sort order to the booking lead-time bands, despite following standard troubleshooting steps.

5. Important Decisions Made and Why

Several decisions were made during implementation:

- I retained the original dataset and performed analysis without overwriting the source data, in line with the project requirements.
- I treated `reminder\_channel` missing values as structural missingness rather than imputing a reminder channel where no reminder was sent.
- I used descriptive and correlational language instead of claiming that any factor causes No-Shows.
- I used 64.1% as the highest observed segment rate rather than presenting it as a prediction or individual patient risk score.
- I kept the dashboard relatively focused and used Appointment Type and Gender as interactive filters for additional exploration.
- I attempted to add a chronological sort for the booking lead-time bands, but encountered a circular dependency error in Power BI that I was unable to resolve within the available time; the bands currently display in default order, which does not affect the accuracy of the underlying figures.

6. Changes to the Week 4 Approach

The Week 4 approach was not fundamentally changed; it was implemented and expanded.

In Week 4, I proposed calculating five KPIs and examining their relationship with appointment outcomes.

During Week 5, I extended this approach by:

- adding exploratory analysis for appointment type and gender;
- performing a combined analysis of booking lead time and previous No-Show history;
- translating validated Python findings into Power BI;
- adding interactive dashboard filters;
- documenting the limitations and considerations of the analysis.

The main change was therefore moving from KPI identification and justification to actual calculation, analysis, visualisation, and interpretation, which is the intended transition for Week 5.

7. Cross-Track Collaboration Completed

A meaningful dependency identified for the wider HealthConnect project is the connection between Data Analytics and Data Science.

The Data Analytics work provides validated KPIs and observed No-Show patterns that can help inform the Data Science track's understanding of potential modelling features and the No-Show problem.

The main analytical outputs shared for this dependency were:

- booking lead time;
- previous No-Show history;
- reminder status and channel;
- distance to clinic;
- appointment outcome patterns.

These findings can provide an evidence-based starting point for further modelling work while making clear that the Week 5 Data Analytics results are descriptive and do not constitute a predictive model.

8. Remaining Work

The main remaining work includes:

- Further refinement and testing of the Power BI dashboard.
- Improving dashboard presentation and usability.
- Further validation of findings where appropriate.
- Preparing the final project documentation and portfolio structure.
- Supporting the next stage of the HealthConnect multidisciplinary solution.
- Incorporating additional evidence or analysis if required by the later project stages.

Week 5 is intended to provide a usable foundation for the next stage rather than complete the entire HealthConnect solution.

9. Proposed Focus for Week 6

For Week 6, my proposed focus is to refine and extend the analytical outputs based on the Week 5 findings.

This will include improving the Power BI dashboard, validating the most important patterns, strengthening the communication of business insights, and collaborating with other HealthConnect tracks where the analytical findings can support their next-stage development.

Overall Week 5 Outcome

Week 5 successfully moved the HealthConnect Data Analytics work from initial planning and KPI identification into practical analysis and dashboard development.

The resulting notebook and Power BI dashboard provide a structured analytical foundation for further HealthConnect development in Week 6.